BSES

POWERING DELHI
EMPOWERING CONSUMERS

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Loss Reduction Journey

BSES: 23 years of serving the National Capital

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BSES Yamuna Power Limited

Powering Delhi, Empowering Consumers

START

THINK

ANALYSIS

PLAN

DO

CHECK

ACT

SUCCESS

LOSS REDUCTION JOURNEY …….

Contents

- Issues- Pre-privatization.
- BSES Profile
- Loss Reduction Journey
- Loss Reduction Initiative
- Change in Strategy
- Achievement

Issues- Pre Privatization

BSES Inherited marred by Multiple Maladies

- Old aged Network
- High Theft
- Outdated Technology
- Poor Consumer Service
- Burn outs Equipment
- Poor resolution Mechanism
- Inadequate Investment
- Unprofessional Practices

Resulted in High Losses & Public Out Cry

Privatization of DVB to BRPL & BYPL in 2002 & thus the initiation of Journey for Improvement

BSES Profile

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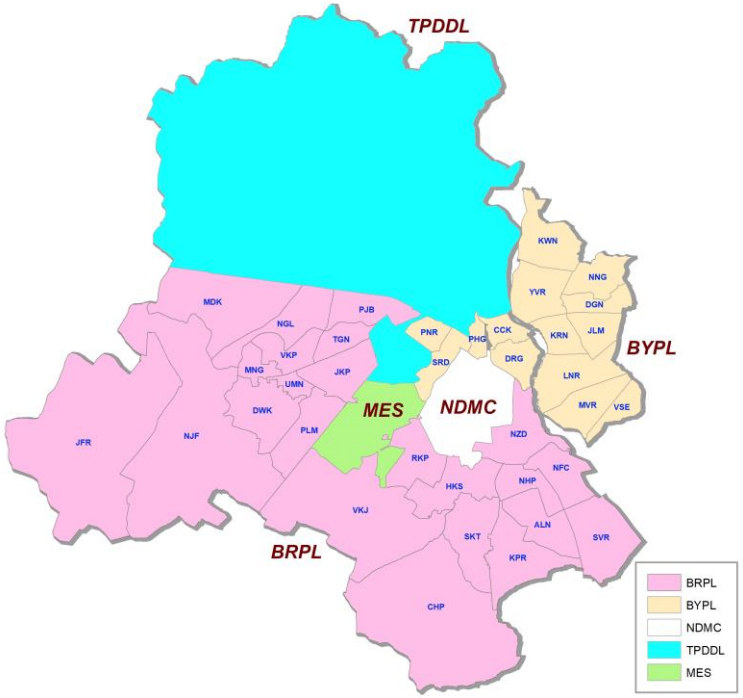
RELIANCE Infrastructure

51 : 49 Joint Venture
of R-Infra & GoNCTD formed on 1st July 2002



Parameter	BYPL	BRPL	BSES
Distribution Area (Sq. km.)	160	691	**851**
Customers (Lakh)	20.4	31.9	**52.3**
Peak Demand (MW)	1,882	3,809	**5,691**
Sales (MU)	7,497	14,706	**22,203**
Sales (Rs Cr)	6,635	12,964	**19,599**
AT&C Loss (%) ^	6.02	6.13	**6.14**

- *Data as of March'25*

^ Provisional; Subject to DERC approval

Delhi peak demand constitutes ~3.5%* of India's Peak Power Demand

** In 2024-25, India's peak demand rose to 249,856 MW in May-24, Delhi's peak demand: 8,656 MW in July-24.*

BSES Customer Profile

Domestic
14,103 MU (63.5%)
4.38 Mn (83.9%)

Commercial
5,517 MU (24.9%)
0.80 Mn (15.3%)

Industrial
998 MU (4.5%)
13.2 k (0.3%)

Agricultural
28 MU (0.1%)
6.4 k (0.1%)

Others
1,557 MU (7.0%)
23.0 k (0.4%)

**** Total Consumer base is over 5.2 Million***

** Data as of March'25*

Steep AT&C Loss reduction post- privatization

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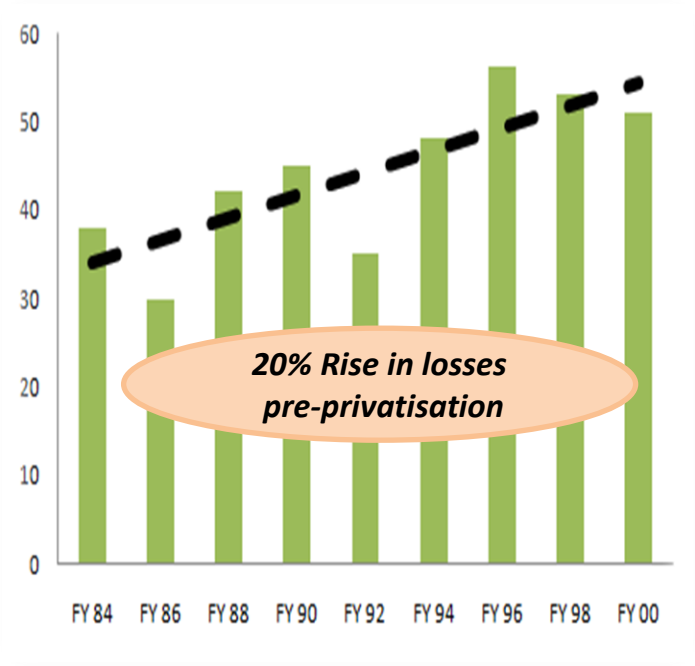
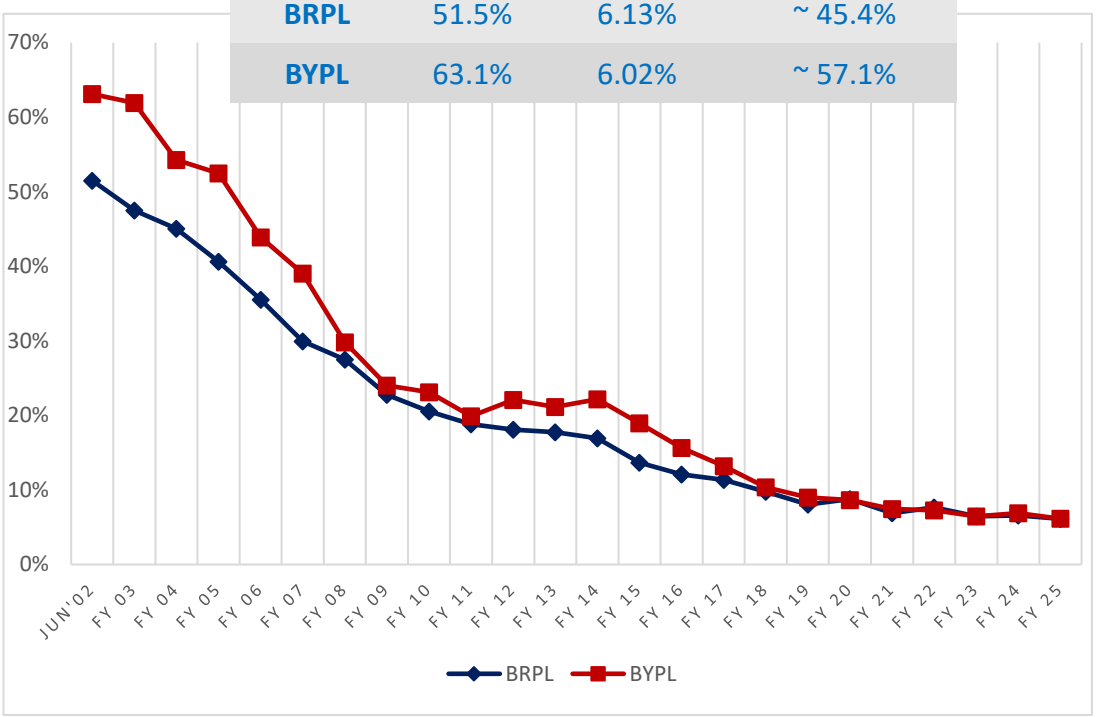
Pre privatisation era



Post privatisation era

Utility	FY 02	FY 25	Reduction
BRPL	51.5%	6.13%	~ 45.4%
BYPL	63.1%	6.02%	~ 57.1%



Around ~ 50% reduction in losses post takeover against rising losses during privatization

*AT&C loss – Aggregate Technical and Commercial loss , * Data as of March'25*

Major Initiatives taken from Post Privatization till FY-14 Phase-I

IT Implementation

During the inception of the company, IT implementation was the priority focus so that data at field can be captured. In order to facilitate the consumer and address the consumer complaint, requests, for robust billing system and improved Power supply system, the need for data capturing was the immediate requirement therefore IT infrastructure was developed.

Setting of IT Infrastructure

- Establish IT infrastructure
- Collecting the entire data and digitising all the data.

Hardware

- Procuring Hardware
- Setting up of Network Connectivity

Software Skills

- Creating User Friendly modules
- Educating Workforce about the software Modules

This initiated capturing of Data from all field for Analysis

IT software which were initially in place were EBS, BAM,NCCM, SAP ISU, DAM & SCADA etc

BYPL Initiatives –Setting up of Customer Friendly Environment

BSES Set Benchmark with **"A+ Ratings in REC's (Under MoP) Nationwide Consumer Service Ratings of Discoms (CSRD)"**, This is the highest ranking for any power distribution company in the country.

Initiation of Meter Modernisation Program

To Arrest under recording of billing Units

Mechanical Meter Changed to Electronic /Static Meter

BYPL was among the first leading utilities to implement 100% Downloading of meters reading.

Initiation of Replacement of Bare Conductor to LT AB

Preventing direct hooking from Bare Conductor

Change of Bare Conductor to AB Cable With DB for Service Line Termination

Initiation of replacement of Old Service Line with Joints and Cut to Armoured Service Lines

Preventing direct hooking from Service Lines

Old & damaged service Cables

<u>Change of Black / Service Line with Cut and Damage to Armoured Cable</u>

Major Initiatives after FY-15- Phase -II

Loss Reduction- Strategy & Planning

Strategy formulation post stagnation & creation of Mission

Management set a mission of Loss reduction in FY-15 as their top most priority.

- *Creation of Project Management Office*
- *Senior Management Guidance on Road map*
- *HR extended their support in restructuring*

Planning at Ground Level

Aligning the entire work force with single objective of loss reduction.

- *Series of workshops, framework development, activity selection and owning the challenge with targets*
- *Creation of Online module for tracking.(DTM)*

Stake-holders alignment for LR plan

This completes the plan :-

- *All the stake holder's departments target was aligned with divisions target.*
- *Only one goal – LOSS REDUCTION*

A Good plan is half the work done

Dedicated Team- Project Management Office

Mission 10%

Bridge between Management & Field Teams

Focus Area

Management established the department for:-

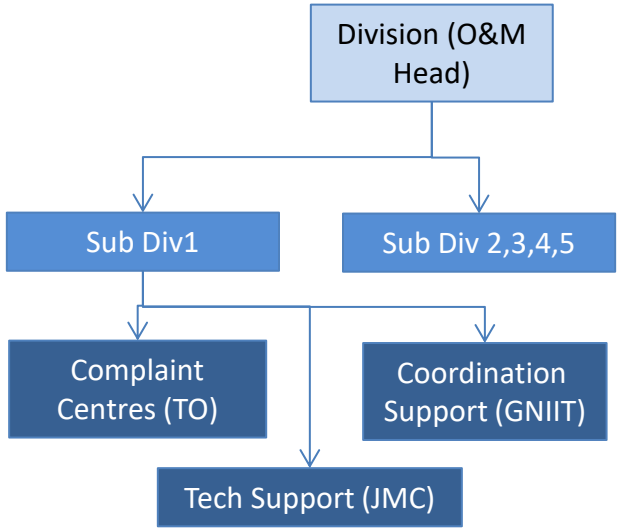
- Creation of Road Map
- Alignment of entire LR Team & stake holders till their last mile.
- Building framework.
- Target Setting.
- Support field team & coordination with stake holder department
- Daily tracking & monitoring the entire L R Project through DTM.
- MIS & all meeting coordination
- Extended arms of management to support the mission.

Outcome of Project Management Office

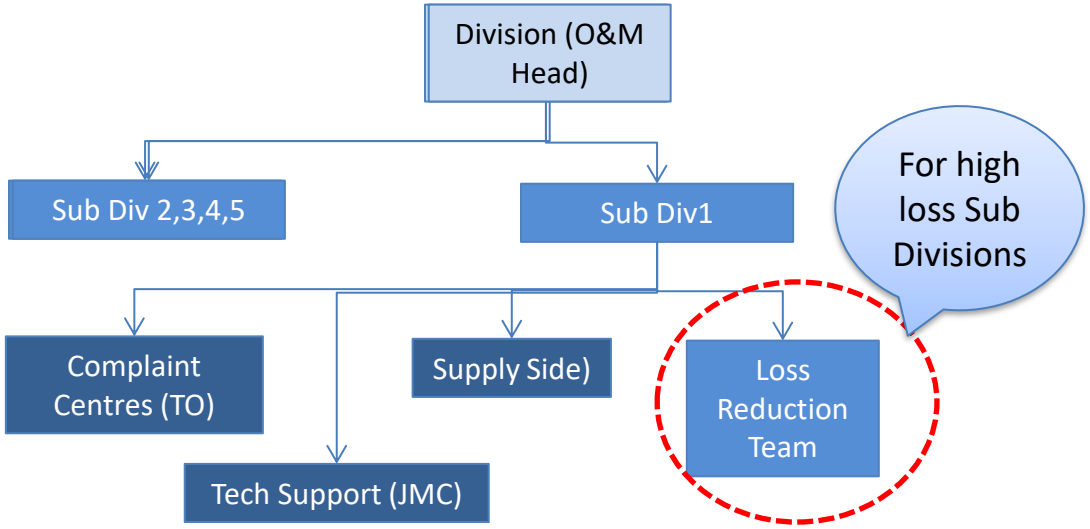
- Top down strategic alignment
- Bottom up execution driven alignment.
- Deliver value added beyond " On time " " In budget".
- Improve visibility & Communication to team & entire management .
- Lead Successful loss reduction journey.

Re-Structuring- Emphasis on LR

Existing Structure



Revised Structure



Additional Structure in the O&M envisaged

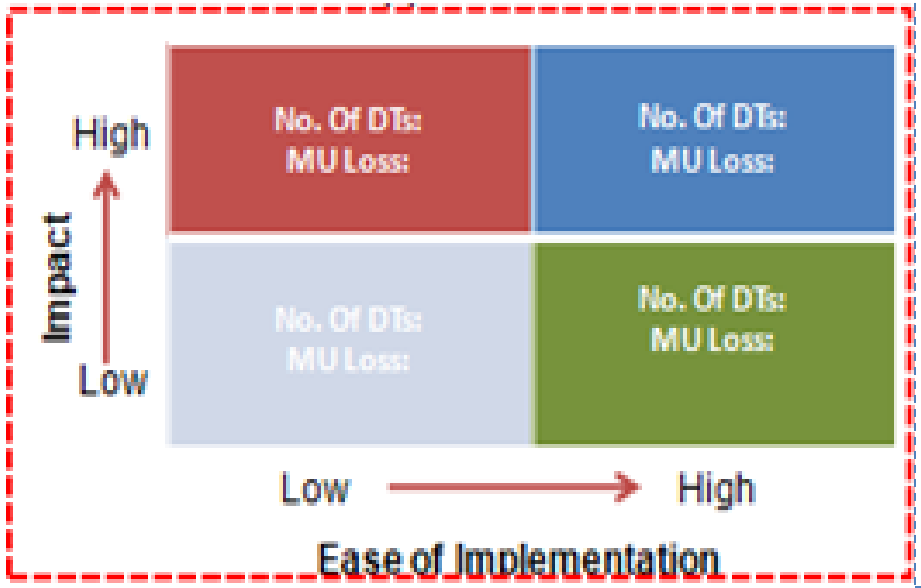
- Right person at right location
- To strengthen loss reduction drive in high loss divisions & Sub divisions.
- Mobilisation of high performing manpower on critical position

Loss Reduction Plan- Strategy

The broad loss level to be achieved is segregated into technical and commercial loss so as to clearly indicate the Divisions loss specially the quantum of commercial loss

Total Loss	HT Technical Loss	
	LT Loss	LT Technical Loss
		LT Commercial Loss

Based on above loss level, reduction in division wise losses various activities were planned and executed which were obtained from workshops.

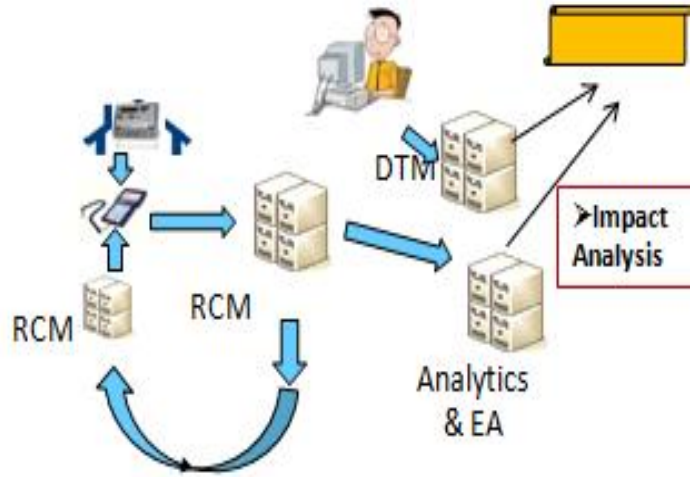


Change in approach was there which helped to reduce the losses by following:-

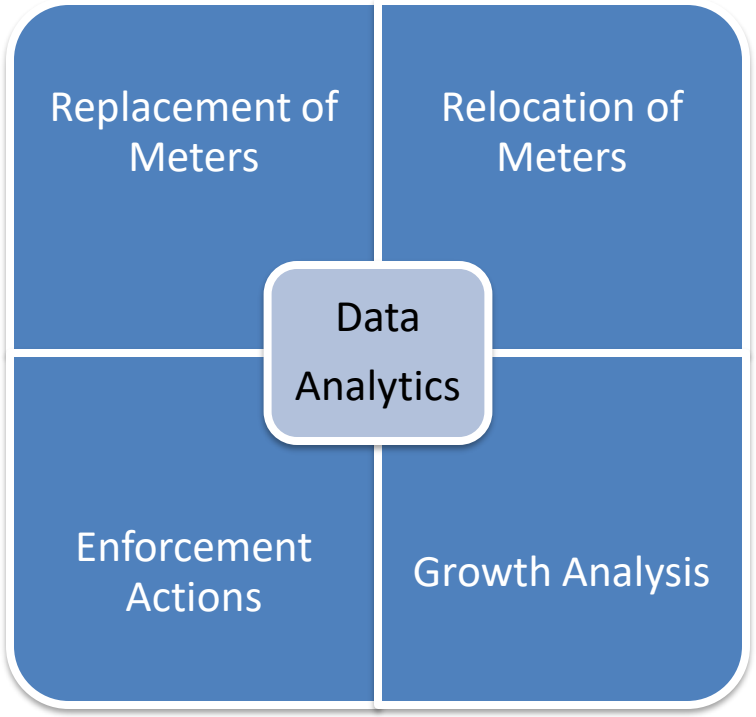
- **Ease of implementation & Impact was prioritized and quadrant formation.**
- **Substation selection was accordingly selected.**
- **Low hanging fruits was focused for immediate gains .**
- **Divisions were also bifurcated as R0,R1 and R2.**
- **Shift from Division to Sub Division level SD0, SD1 & SD2 and prioritizing High loss .**

Workshop Planning- Bottom Up Approach

Data Analytics from Various Sources



Analytics Department was restructured, who was responsible for complete Data Analysis from various software tools.



Creation of DT Tracking Module (DTM)-

- Framework uploaded in DTM.
- To track the entire activities.
- Stake holder dept also updated leads in DTM
- Closure was also monitored through this which helped for complete monitoring through same field.
- For such a huge data handling, MIS monitoring , DTM played vital role.
- Assisted field team , supervisors and management for proper assessment

Strategic Initiative:- Meter Related Activities

Meter Relocation

Outside the premises

Meter Revamping

Pole Metering by installation of IGMS

Strategic Initiative-Network Related

LT AB PUNCTURING Which is being Replaced by Armoured

Interventions taken to avoid Cable Insertion

Locking of DB

Welding of DB

Compounding of DB

Creation - Surveillance Department

Separate Structure for Surveillance

❖ For independent & transparent observations.

❖They supported both LR team & enforcement team by continuous infusion of leads.

❖ Surveillance acted as eyes of management for checks & balances at site.

Surveillance department :- WHY

❖ Loss reduction required sustenance.

❖Provide field input post & pre work done

❖ All the observations acted as a lead.

❖Odd Hours surveillance

❖Presence in field 24x7

❖Deterrence in field

Touch-Enforcement & Soft Initiative-Consumer Engagement Program & Media Campaign

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Tough Initiative

During the Journey of Loss reduction, Enforcement Actions were also carried out which created deterrence in consumer for theft of electricity

Soft Initiative

Media Campaign

Both Tough & Soft Initiatives helped in Controlling theft and educating consumer

Strategic Initiative- Assimilating Inputs, MIS(Creation of DTM) & Field Actions

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SUPPORT TOOL

- Energy Audit of the DTs
- Created DT Tracking module to monitor various LR activities
- Analytics Feedback for to Identify Suspect
- Capturing exceptions and Acting on it with downloaded data
- Field Survey

- Enforcement Action
- Surveillance Inputs
- Legal Actions
- Quality Inspection
- Actions from Police Liaison team

FIELD ACTIONS

Field Out Put

Activities
Network Related Activities
Removal of Illegal tapings
Replacement of AB to Armored Cable
Meter Relocation (outside the premises)
Meter Replacement
Theft Booking
Complaints Filed
FIR Lodged

Customer Engagement Program

Activity Name
Sajha Prayas
Pragati
Energy Conservation School drive
RWA Meeting
Nukkad Natak
Jagrukta Rally
Campaign Against Theft
Vartalaap- Virtual RWA meet
Fire & Safety Awareness programs

Close Monitoring of all Above at all Hierarchy

Other Initiatives

- Score Card Implementations at Sub division, Division and for individual performance.
- Score card for stake holder department.
- Periodic R&R programs to motivate the field teams.
- Individual appreciation by CEO , time to time for exhibiting exemplary performance
- Outbound training for field team for learning & motivation.
- Loss reduction incentives on the basis of achievement.
- Vendor Score Card.

Tracking Tools

❑**Activity Monitoring**:- DTM (DT tracking module) a tool is developed in order to monitor the daily activities which are carried out in field.

Associated Activities:-
*LR Initiatives such as Pole & Meter related activities
*Enforcement Update
*Legal tracking
*Surveillance leads capturing & Closure

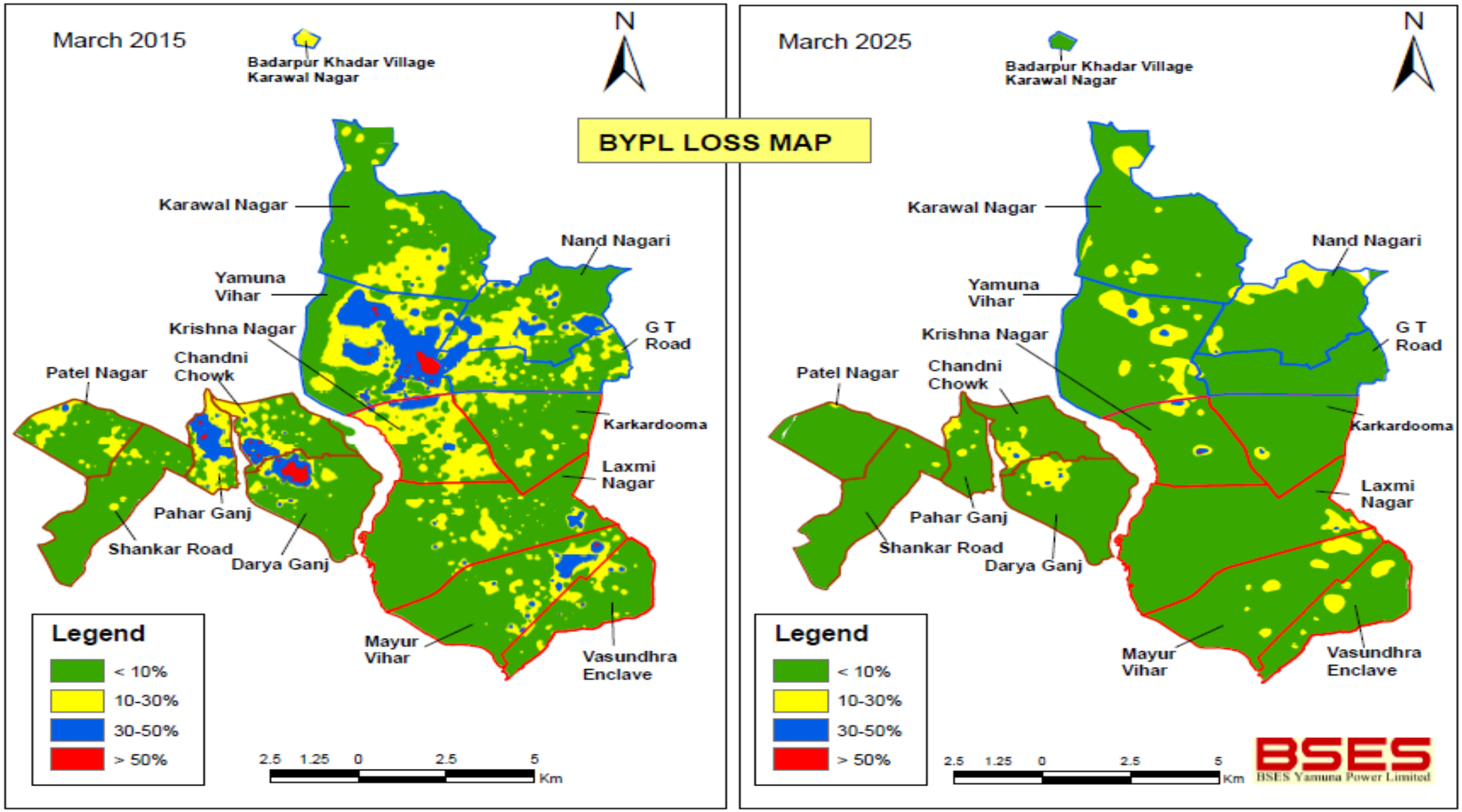
❑**Loss Monitoring** :- Energy audit report reveals the actual result of the substations in which the actions are performed.

❑**Analytics Feedback**:- Analytics helps to analyze the growth of the activities performed at site.

❑ **Feedback & Discussion on laggards:-** During the various reviews at Circle, that is second and fourth Tuesday's meeting at respective circles for division and first and third Tuesday's review for high loss divisions.

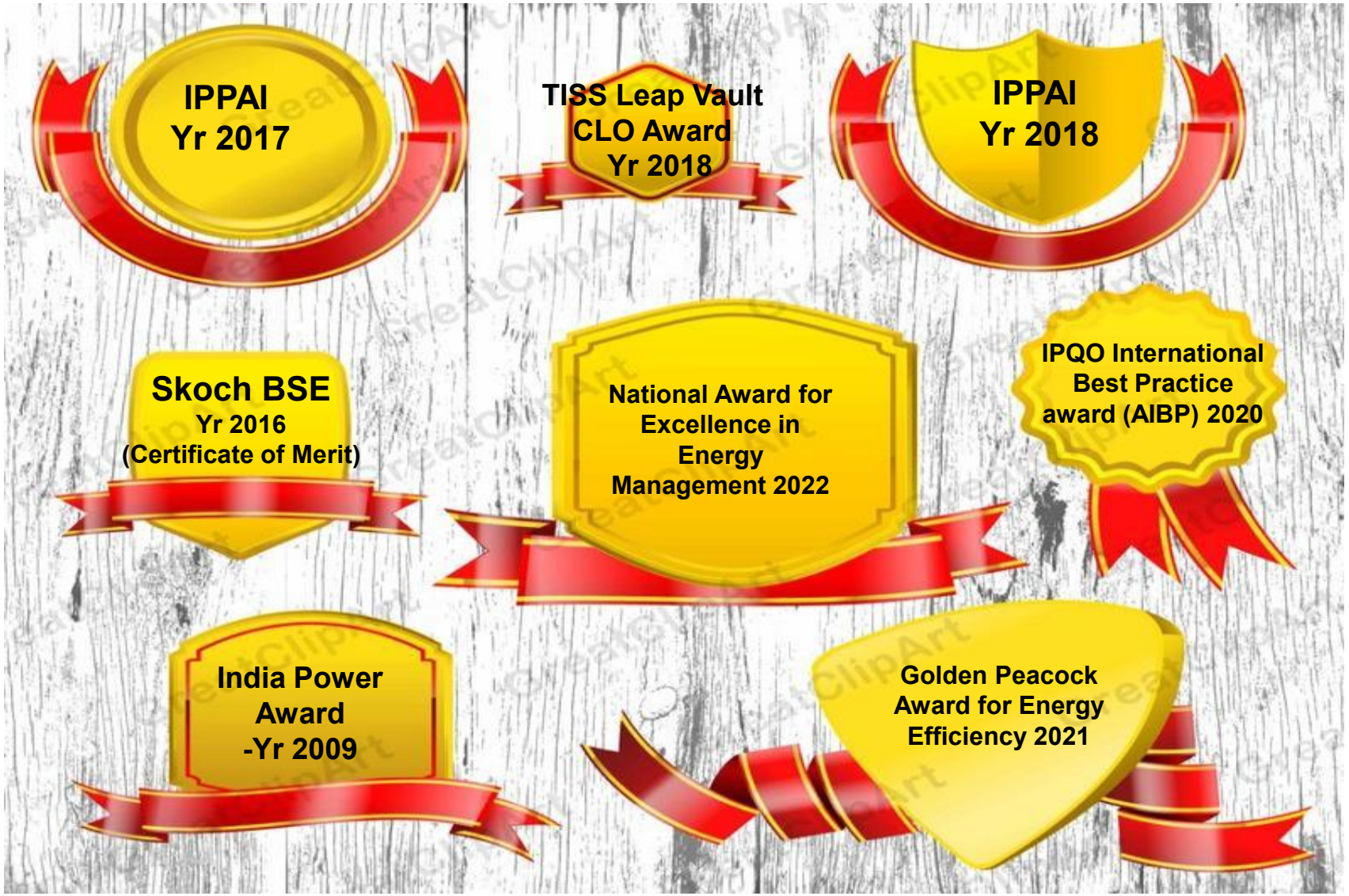
RESULT

Loss Reduction Map - BYPL



Heat map showing visible loss reduction in divisions in recent years

‘Loss Reduction’ Wall of Fame



THANKS